Learning to be rational or irrational ? Explore the decision consistency between human beings and AI Agent in choice behaviour

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Abstract

Since the advent of Large Language Models (LLMs), Generative AI has been extensively applied in the marketing field to improve decision-making efficiency and automate content generation. However, the effectiveness of AI Agent in assisting or substituting for human decisions remains controversial. This research conducted two experiments and trained Generative AI to perform decision-making tasks with different option- framing, aiming to evaluate the decision consistency between LLMs and humans. Results reveal that while LLMs can align with human decision patterns under certain conditions, they also exhibit irrational decision biases.

Materials & Methods

We conducted a 2×2 factorial experiment with GPT-3.5, GPT-4, and 140 human participants, comparing additive vs. subtractive option framing in selecting 12 in-flight services (hedonic/utilitarian). AI agents were prompted with structured consumer traits. Outputs were analyzed for decision time, number of services chosen, and preference type. Statistical analyses included t-tests, Mann–Whitney U tests, and Shannon entropy.

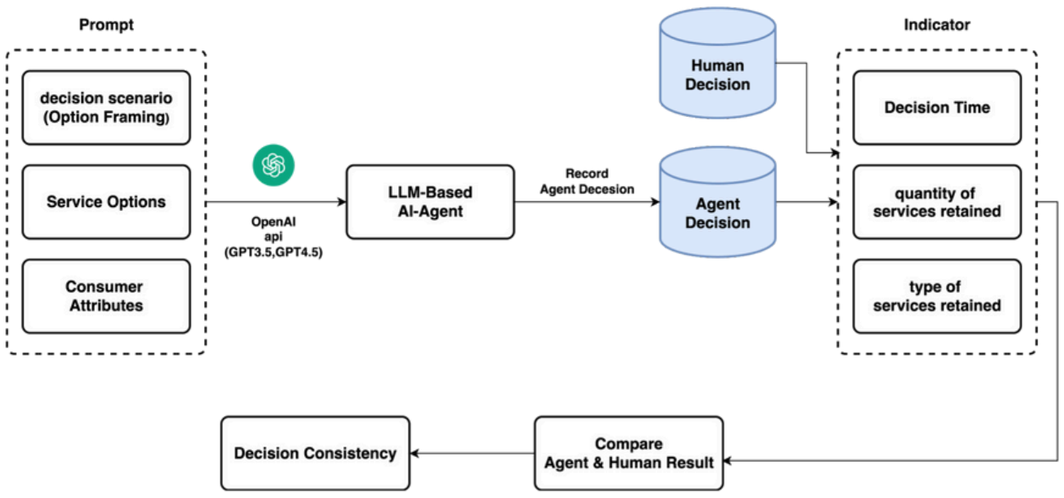


Fig1 Experimental Design/Procedure

Results

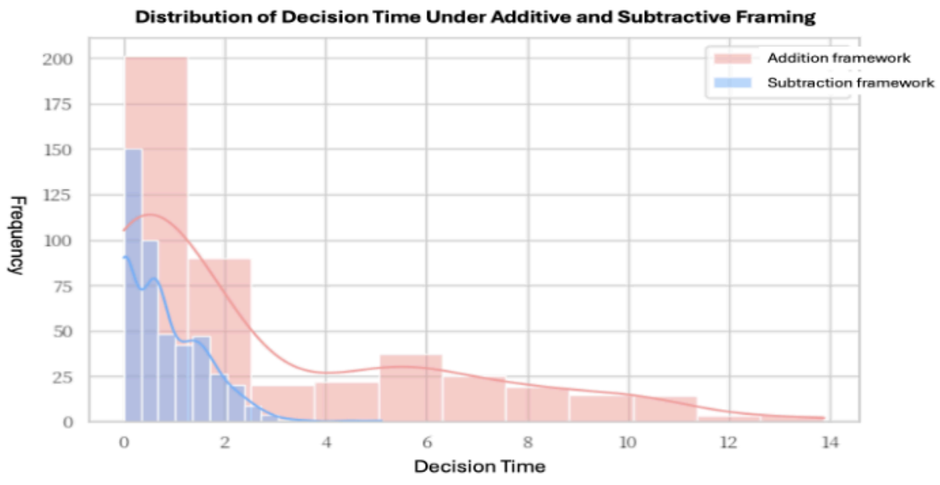


Fig2 Decision Time Comparsion between Framework

Additive framing led to significantly longer AI decision times than subtractive framing.(t = 12.56, p < .001***)

In Fig3, AI agents exhibited a strong status quo bias: under additive framing, 62.9% selected only 3–4 services, while under subtractive framing, 76.7% retained over 5 (t = –21.86, p < .001). Across both framings, agents showed a mild preference for hedonic services (51.5% additive; 54.2% subtractive), indicating limited framing effect on preference type.

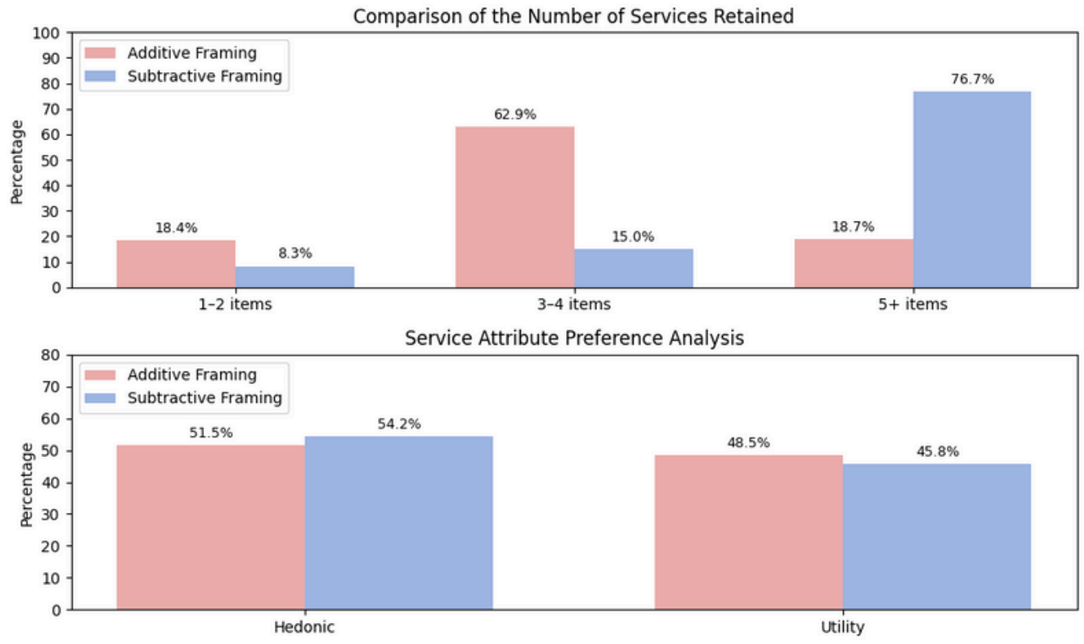


Fig3 Service Retained & Preference between Framing Effect

Tables 1 and 2 show that GPT models deviate from random selection under additive framing, with GPT-4 showing the strongest bias in subtractive framing (lowest entropy = 3.33). GPT-4 aligns more closely with human choices (mean diff = –0.58, p = 0.007), while GPT-3.5 tends to overselect (–2.25, p = 0.09).

Tabel 1 Decision Bias and Fairness Analysis of AI Models in Option Framing

Framing/Model	GPT-4	GPT-3.5	Random
Additive Framing	3.3395	3.2962	3.5749
Subtractive Framing	3.3261	3.474	3.584

Tabel 2 Consistency between AI models and humans in choosing outcomes

	GPT-4	GPT-3.5
Mean	-0.58	-2.25
Standard Deviation	0.37	0.43
Sample Size (n)	150	150
t-value	11.22	3.01
p-value (two-tailed)	**0.007*** **	0.09

Discussion

1. **Framing Effects** : GPT models, like humans, exhibit framing effects. Under subtractive framing, they retain more services and favor hedonic options, reflecting status quo bias and emotional salience.
2. **Model Differences** : GPT-3.5 shows stronger bias under additive framing, while GPT-4 shows more under subtractive framing. Bias may grow with contextual sensitivity, not always decline with model upgrades.
3. **Human-AI Consistency** : GPT-4 aligns more with human choices under additive framing, though both models show unstable consistency in complex contexts.

Conclusion

GPT models reflect both rationality and bias from human data. GPT-4 aligns with humans under additive framing but shows hedonic bias subtractively. Bias grows with context sensitivity. Fairness requires careful prompt framing—additive formats and standardized prompts are recommended for safer AI decisions.